

Local LLM + MATLAB Agentic Toolkit Setup Guide

LM Studio – MATLAB MCP Server – MATLAB & Simulink Agentic Toolkits

Platform: Windows | MATLAB R2026a | **Runs Fully Offline** | Revised 19 August 2026

Overview

This guide sets up a **fully local AI assistant for MATLAB and Simulink**. The language model runs on your own machine, so your code and models never leave your computer.

★ Note

Once the setup is done, the whole thing runs offline. The model, the MCP server, the tools and the skills all live on your own disk — nothing is sent to a cloud service, and no internet connection is needed to use it. You only need to be online for the **downloads** in Steps 1, 3, 4, 5 and 10.

Four pieces work together:

Component	What it does
LM Studio (Bionic)	Runs the language model locally and hosts the chat / agent interface.
MATLAB MCP Server	Lets the agent drive MATLAB, starting it if it is not already open.
MATLAB Agentic Toolkit	Adds MATLAB tools and skills the agent can use.
Simulink Agentic Toolkit	Adds Simulink, Stateflow and System Composer tools and skills.

★ Note

Set aside about **1–2 hours**, most of which is downloading. You will need roughly **15 GB of free disk space** for LM Studio, one model, and the toolkits.

What you will need:

- A **Windows** PC where you are allowed to install software
- **MATLAB R2026a** installed locally, plus **Simulink** for the Simulink toolkit
- A stable internet connection — for the setup downloads only
- A GPU with 8 GB or more of VRAM is recommended, but not required

★ Note

Please complete this setup **at least one day before** the workshop. If you get stuck, contact matlabday_support@techsource-asia.com with a screenshot of the step that failed.

Step 1. Install LM Studio

1. Go to <https://lmstudio.ai/download> and click **Download LM Studio Bionic**. The installer is about 600 MB.

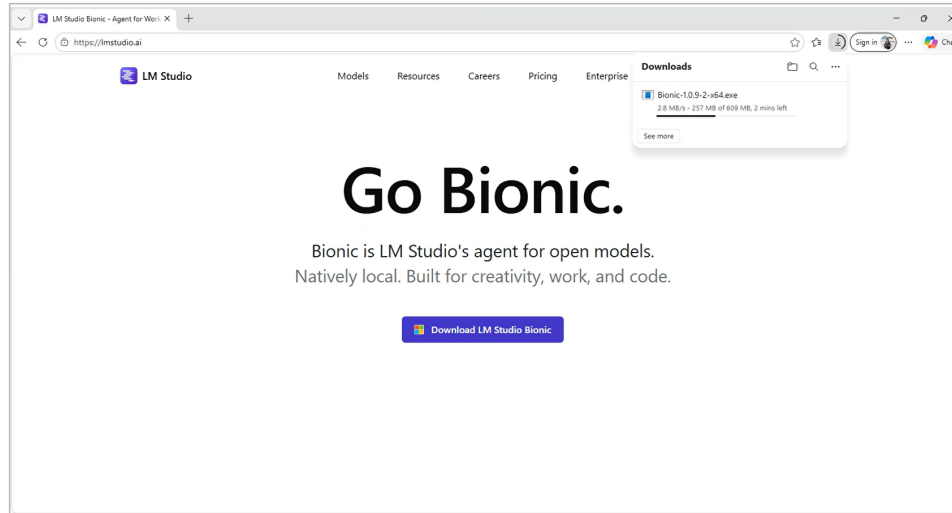


Figure 1: Downloading the LM Studio Bionic installer

2. Run the downloaded `Bionic-version-x64.exe` and accept the default options.

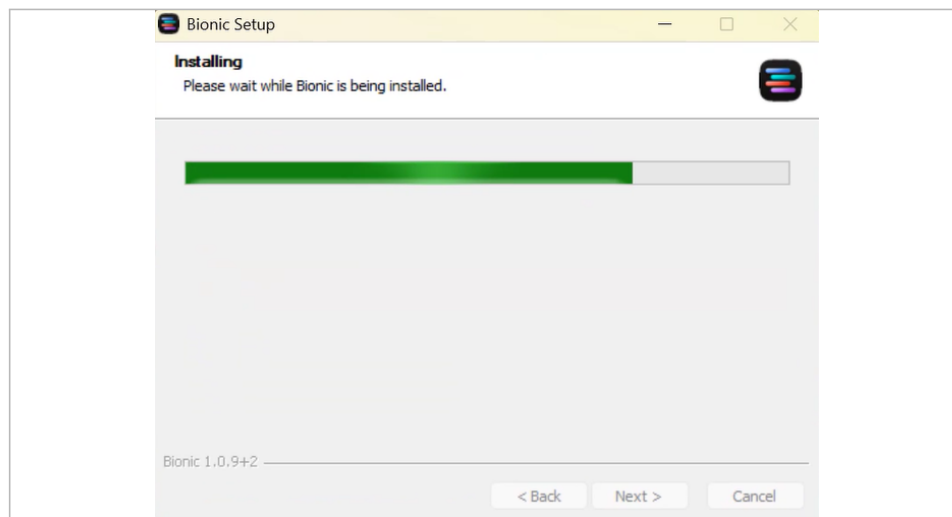


Figure 2: The LM Studio (Bionic) installer running

★ Note

Note down the installation path shown by the installer. Separately, LM Studio keeps your per-user data — models, skills and settings — in a hidden folder in your user profile:

```
C:\Users\<YourUserName>\.lmstudio
```

You will need that second path in **Step 9**.

Step 2. Open the LM Studio Settings

1. Launch LM Studio. On a fresh install you land on the **Create your first project** screen. Click the **sidebar toggle** in the top-left corner.

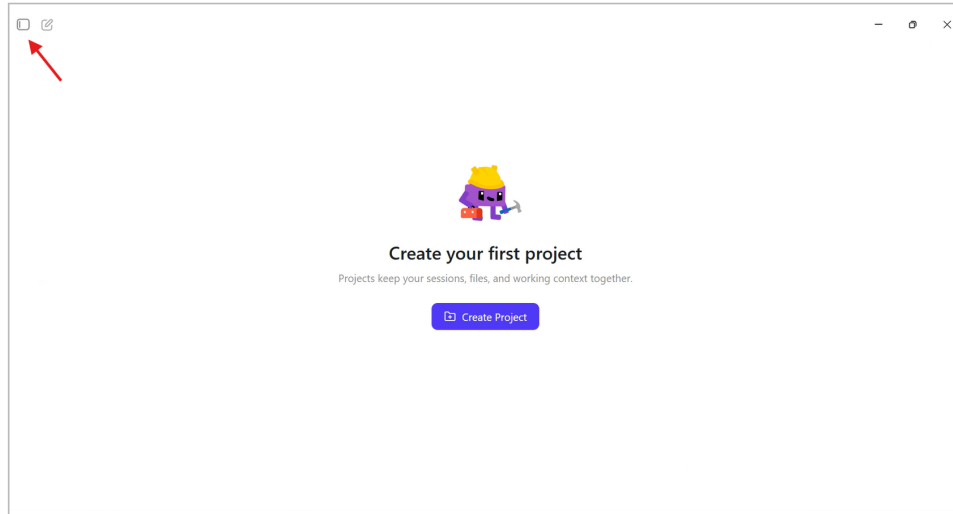


Figure 3: Click the sidebar toggle at the top-left

2. Click **Settings** at the bottom of the sidebar.

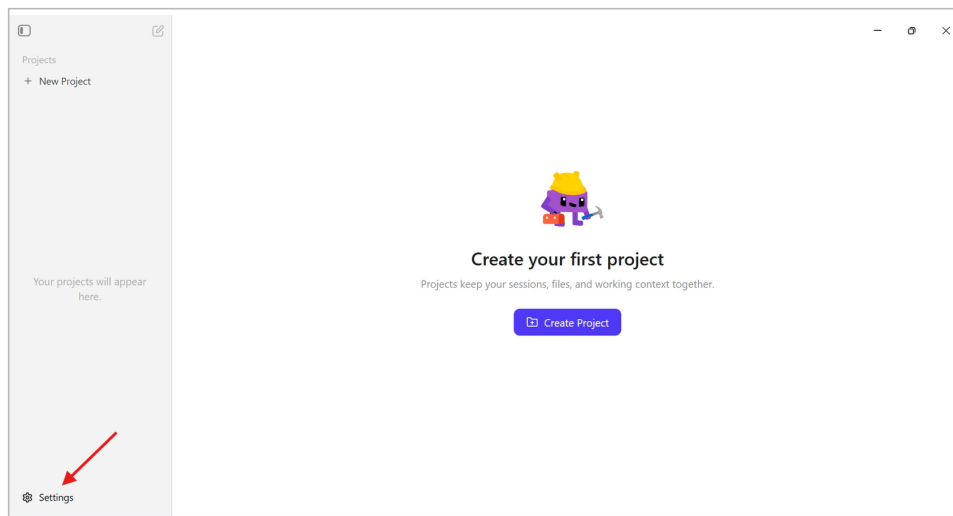


Figure 4: Settings sits at the bottom of the expanded sidebar

Step 3. Choose and Download a Model

1. In the Settings sidebar, under **Local Models**, click **Explore**.
2. Pick a model from the staff picks, or search Hugging Face. This guide uses **Gemma 4 12B QAT** (7.15 GB), which fits comfortably on most workshop machines.
3. Click **Download** and let it finish before continuing.

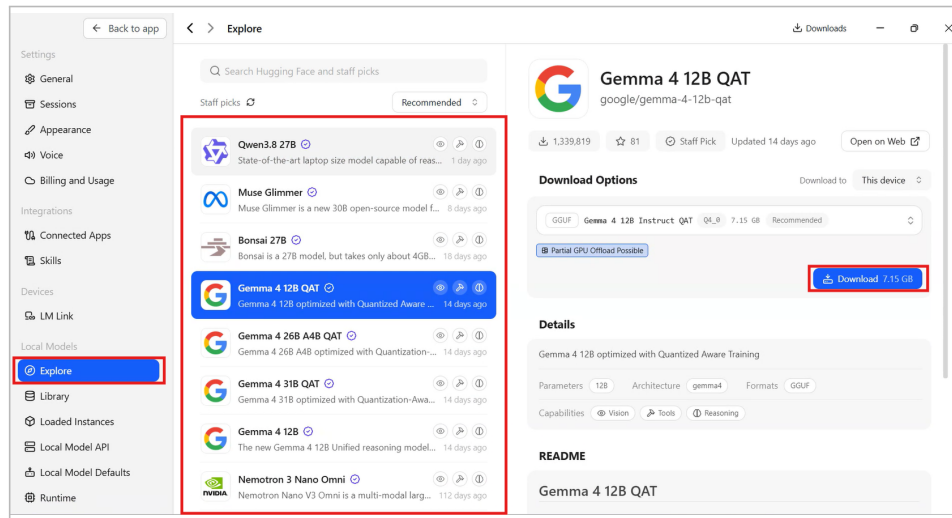


Figure 5: Selecting Gemma 4 12B QAT in Explore and starting the download

★ Note

Check the **Capabilities** row in the model details before downloading. The model **must list Tools** — without it the model cannot call the MATLAB MCP tools and nothing later in this guide will work. Vision and Reasoning are useful extras.

Step 4. Download the Agentic Toolkit Installer

1. Go to the Simulink Agentic Toolkit repository:
<https://github.com/matlab/simulink-agentic-toolkit>
2. In the right-hand column under **Releases**, click the release marked **Latest**.

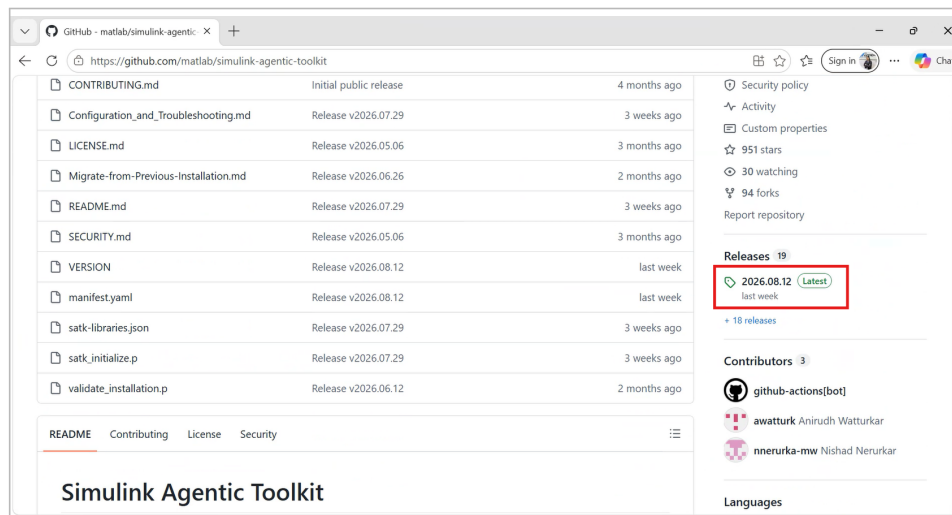


Figure 6: Open the latest release from the repository page

3. Scroll down to **Assets** and download `agenticToolkitInstaller.mltbx`.

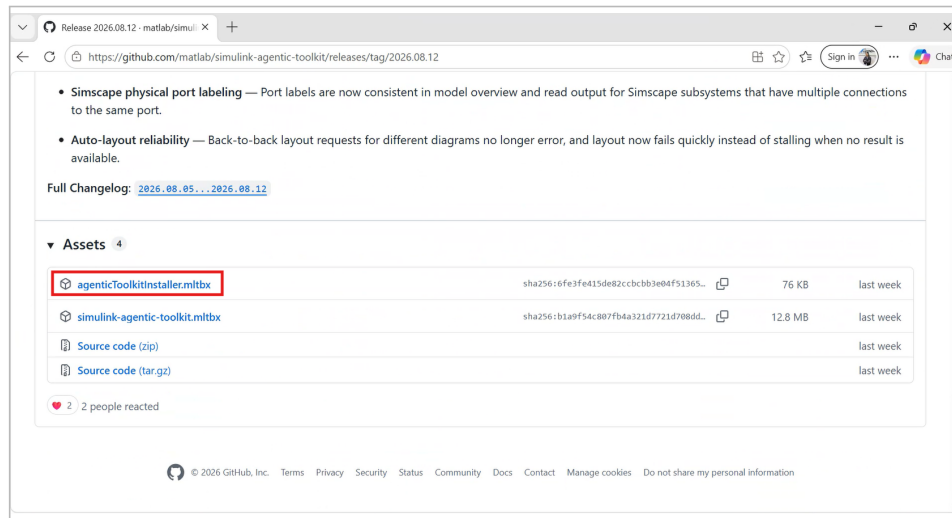


Figure 7: Download agenticToolkitInstaller.mltbx from the release assets

★ Note

agenticToolkitInstaller.mltbx is the **only** file you need. It is a small installer, about 76 KB, that fetches the MCP server and both toolkits for you. Do **not** download simulink-agentic-toolkit.mltbx separately.

Step 5. Install the Toolkits in MATLAB

1. Open MATLAB and set the **Current Folder** to the folder you downloaded the installer into, for example Downloads.
2. In the **Command Window**, run the following, substituting your own path:

```
matlab.internal.open.openmltbx("C:\Users\TSA\Downloads\agenticToolkitInstaller.mltbx")
```

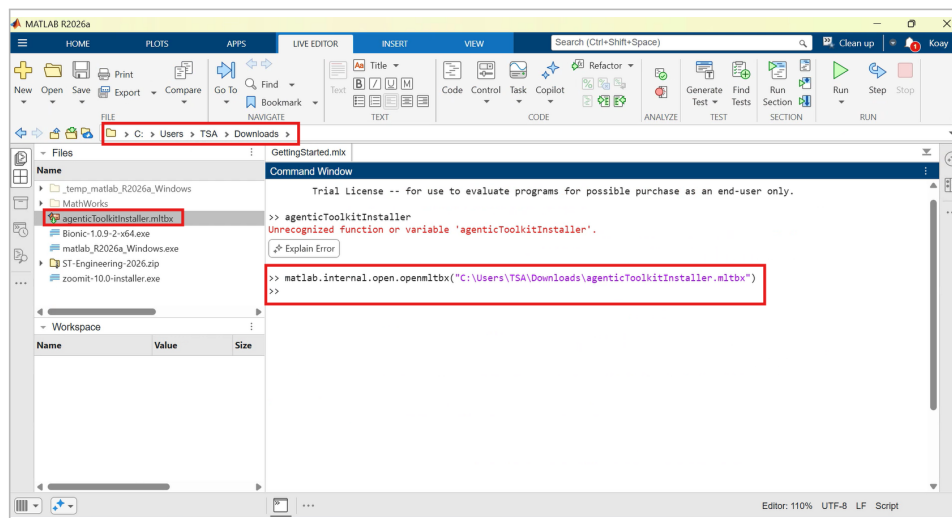


Figure 8: Running openmltbx on the downloaded .mltbx file

3. When **Installation Complete** appears, click **Close**.

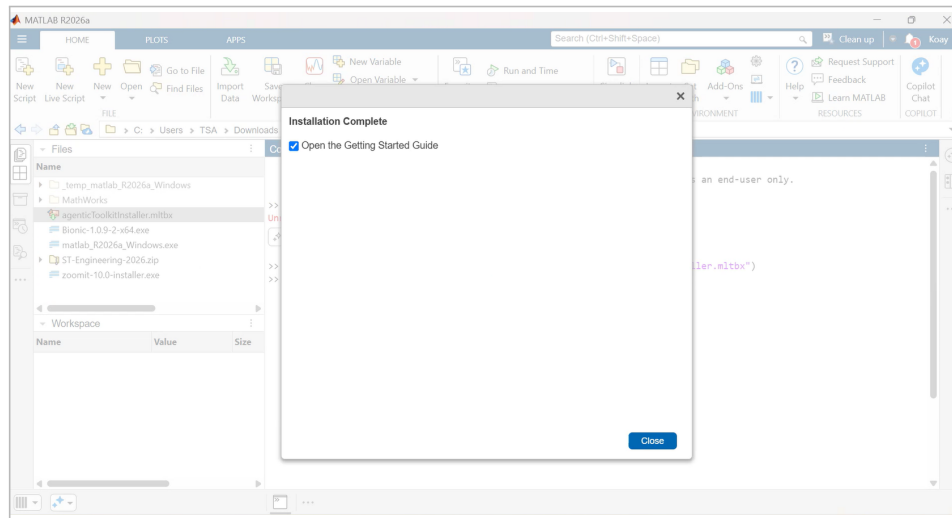


Figure 9: The installer confirms the toolbox is installed

4. Now run the setup function:

```
setupAgenticToolkit("install", DisableTelemetry=true)
```

5. At **Enter selection**, type 3 for **Both** and press Enter. Review the plan, then type **y** at **Proceed? [Y/n]** and press Enter.

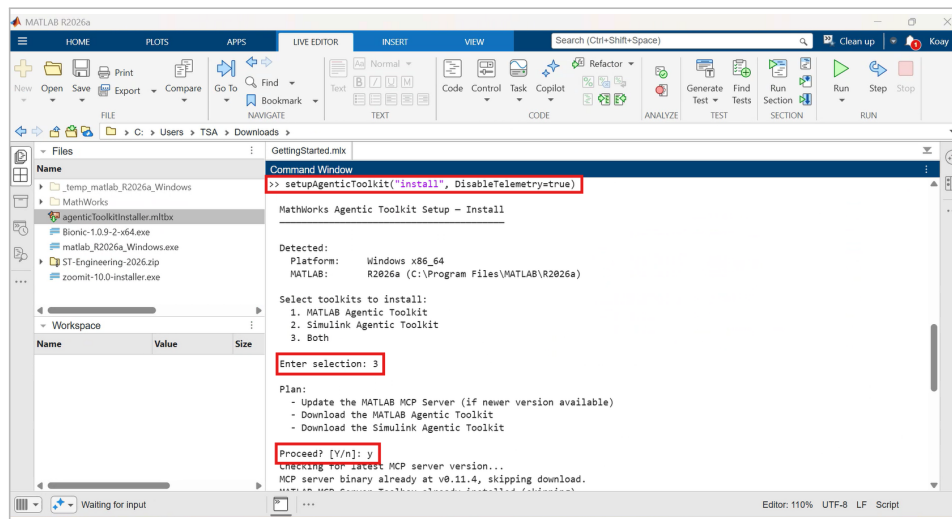


Figure 10: Select option 3 (Both), then confirm with y

6. The setup downloads and installs the MCP server and both toolkits. When it asks “**Would you like to configure your coding agent now?**”, type **n** and press Enter.

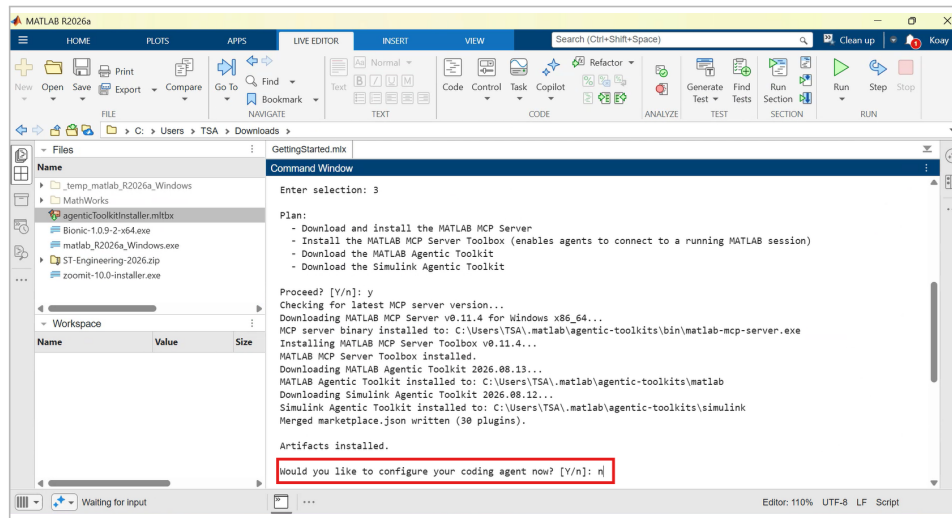


Figure 11: Answer n — LM Studio is registered by hand in Step 7

★ Note

Answer n because the automatic configuration only covers Claude Code, Codex, GitHub Copilot, Amp and Gemini CLI. LM Studio is registered manually in **Step 7**.

Step 6. Copy the MATLAB MCP Server Path

1. In the setup output, find the line beginning **“MCP server binary installed to:”** and select and copy the full path.

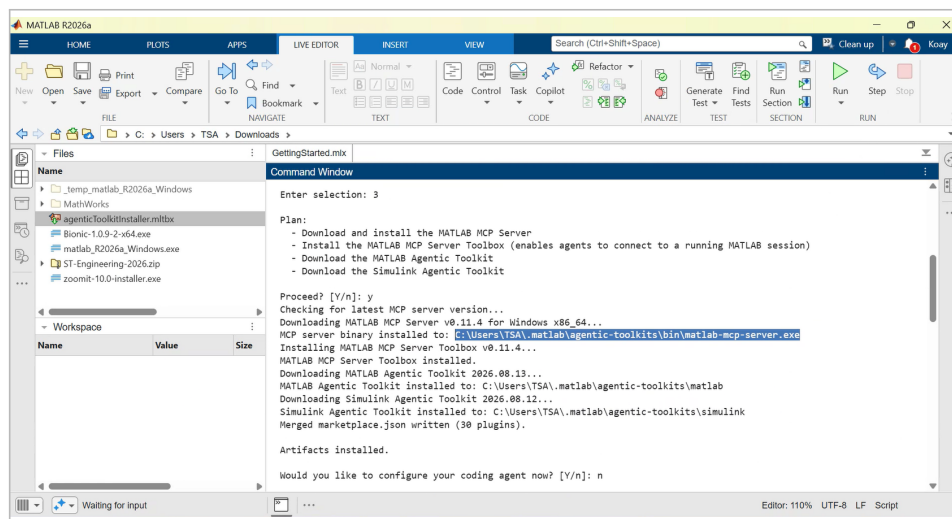


Figure 12: Copy the MCP server binary path from the install log

On a default installation the path looks like this:

C:\Users\TSA\.matlab\agentic-toolkits\bin\matlab-mcp-server.exe

★ Note

Leave the MATLAB Command Window open and do not clear it. You will come back for a second path in the middle of the next step.

Step 7. Register the MCP Server in LM Studio

1. In LM Studio, go to **Settings** → **Connected Apps**. At the bottom, under **Manual MCP server setup**, click **+ Add custom MCP**.

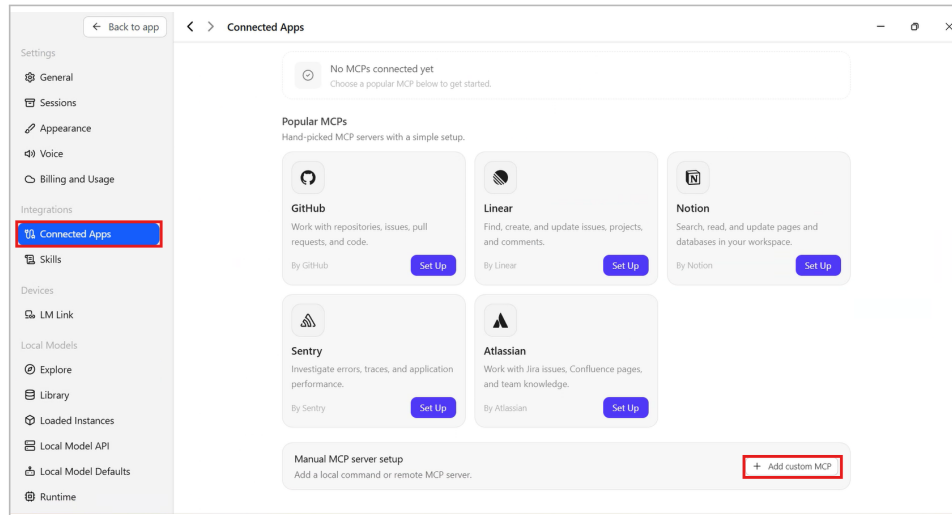


Figure 13: Add custom MCP under Manual MCP server setup

2. Fill in the form:

- **Name:** MATLAB MCP Core Server
- **Connection:** On this computer
- **Command:** paste the MCP server path you copied in Step 6

3. Switch back to MATLAB and copy the **Simulink Agentic Toolkit** path from the line “**Simulink Agentic Toolkit installed to:**”.

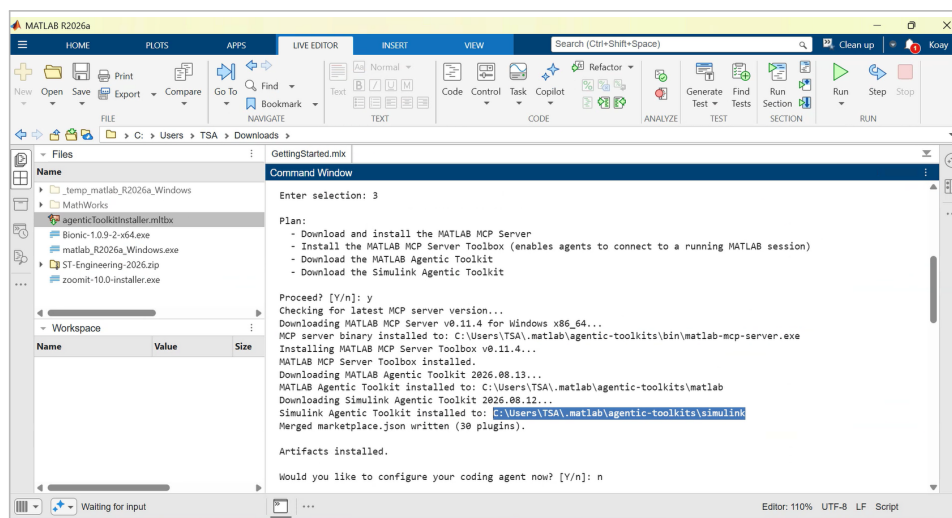


Figure 14: Copy the Simulink Agentic Toolkit installation path

4. Back in LM Studio, turn on **Show advanced options**. Under **Arguments**, add one argument built from the path you just copied:

```
--extension-file=C:\Users\TSA\.matlab\agentic-toolkits\simulink\tools\tools.json
```

Replace `C:\Users\TSA\.matlab\agentic-toolkits\simulink` with your own Simulink Agentic Toolkit path, and keep the trailing `\tools\tools.json` exactly as shown.

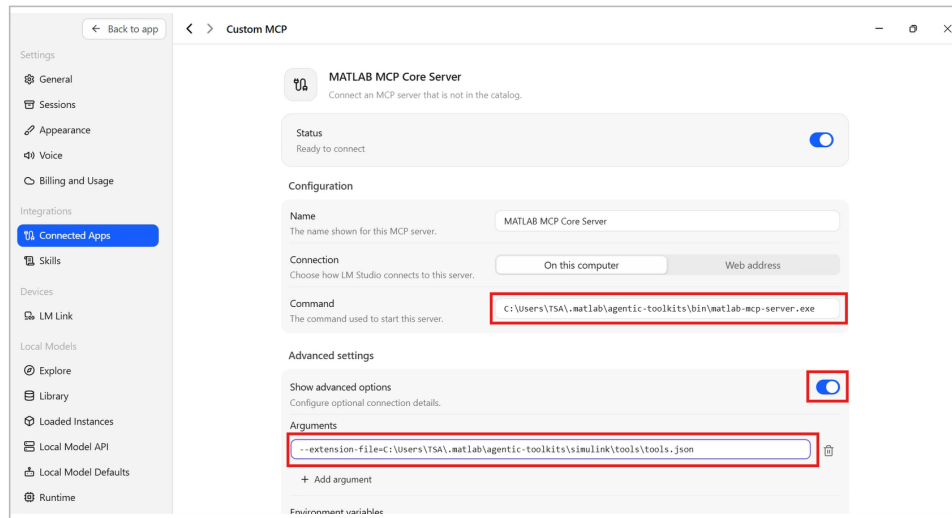


Figure 15: The Command field and the extension-file argument

5. Scroll to the bottom of the form and click **Add MCP**.

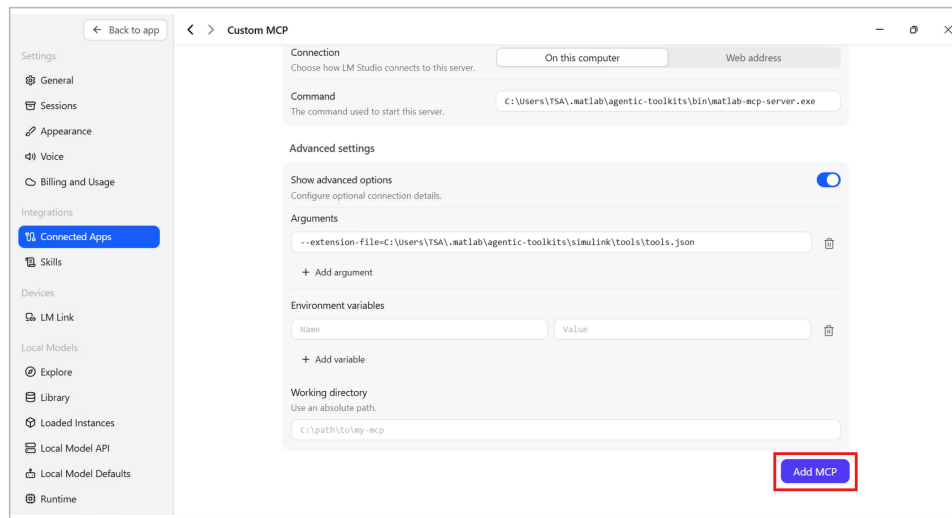


Figure 16: Click Add MCP to save the server

6. Back on the **Connected Apps** page, confirm the server reports **Connected • 13 tools ready**.

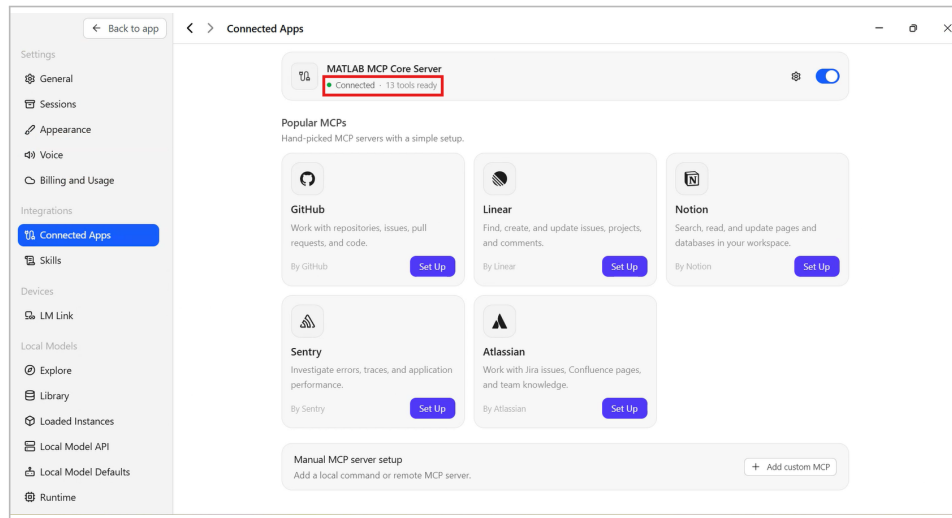


Figure 17: The MATLAB MCP Core Server connected with 13 tools

★ Note

If you do not see **13 tools ready**, check these in order:

- Toggle the server **off and on again** — LM Studio only launches the server when it is enabled.
- The **Command** path points at `matlab-mcp-server.exe` and that file exists.
- The `-extension-file` argument ends in `\tools\tools.json` and that file exists. Without this argument you will see fewer tools, because the Simulink tools are never loaded.

Step 8. Increase the Context Length

1. In Settings, under **Local Models**, click **Local Model Defaults**.
2. Set **Minimum AutoFit context length** to 325000.

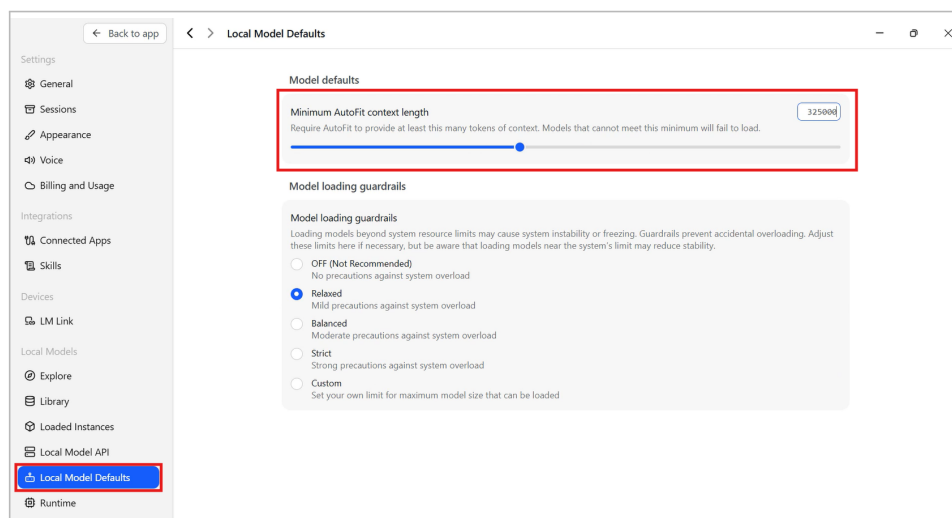


Figure 18: Raise Minimum AutoFit context length to 325000

★ Note

The toolkit tool definitions and skills are large. With too little context the agent silently truncates them and tool calls start to fail. If a model then **refuses to load**, your machine cannot provide that much context — lower this value or choose a smaller model.

Step 9. Open the LM Studio Skills Folder

1. Open File Explorer, paste the following into the address bar, and press Enter:

```
%USERPROFILE%\lmstudio\skills
```

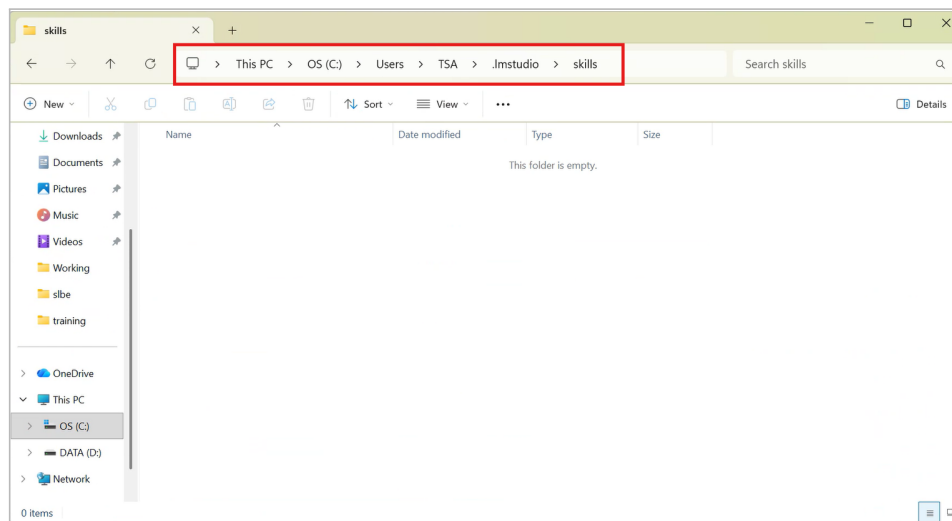


Figure 19: The hidden .lmstudio skills folder, empty on a fresh install

2. Leave this window open — you will paste into it in the next step.

★ Note

.lmstudio is hidden, so you will not find it by browsing. Paste the path above, or turn on **View** → **Show** → **Hidden items**. If the `skills` folder does not exist yet, create it.

Step 10. Install the MATLAB and Simulink Skills

1. Open both toolkit repositories in your browser:

- <https://github.com/matlab/matlab-agentic-toolkit/>
- <https://github.com/matlab/simulink-agentic-toolkit/>

2. On each repository, click the green **Code** button and choose **Download ZIP**.

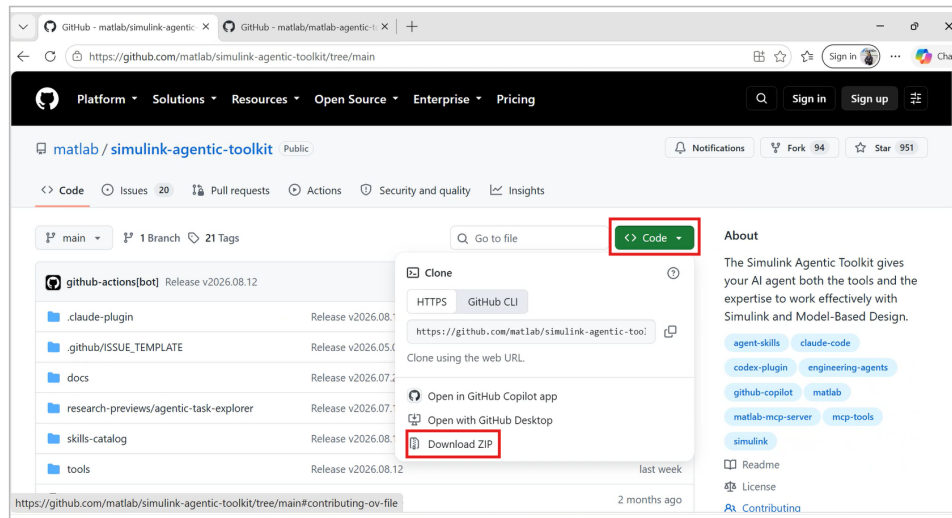


Figure 20: Code, then Download ZIP, on each repository

3. Extract both ZIP files. Open matlab-agentic-toolkit-main\skills-catalog, select **all of the folders**, and copy them.

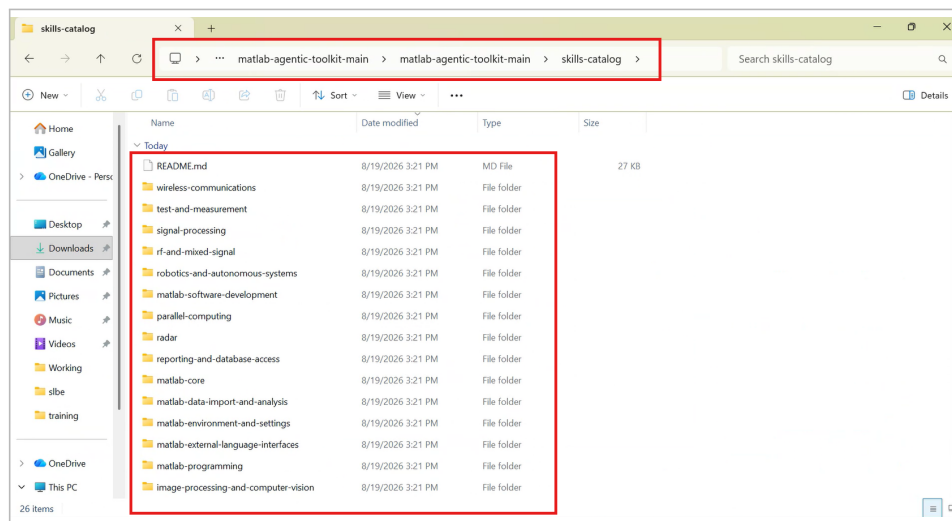


Figure 21: The MATLAB skills catalog — copy all folders

4. Paste them into the .lmstudio\skills folder from Step 9.
5. Repeat for the Simulink toolkit: open simulink-agentic-toolkit-main\skills-catalog, copy all of the folders, and paste them into the same skills folder.

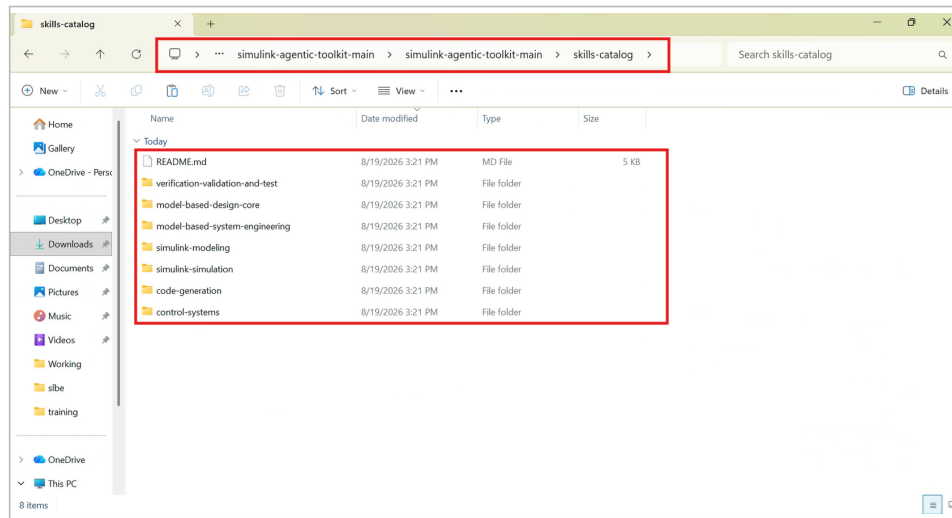


Figure 22: The Simulink skills catalog — copy all folders

★ Note

Copy the **contents** of skills-catalog, not the skills-catalog folder itself. Each skill category must sit directly inside .lmstudio\skills. You can skip README.md.

Step 11. Verify the Skills Are Loaded

1. In LM Studio, go to **Settings** → **Skills**.
2. Confirm the **Bionic Skills** list is populated, with entries such as building-simulink-models, checking-model-compliance and authoring-simulink-inputs.

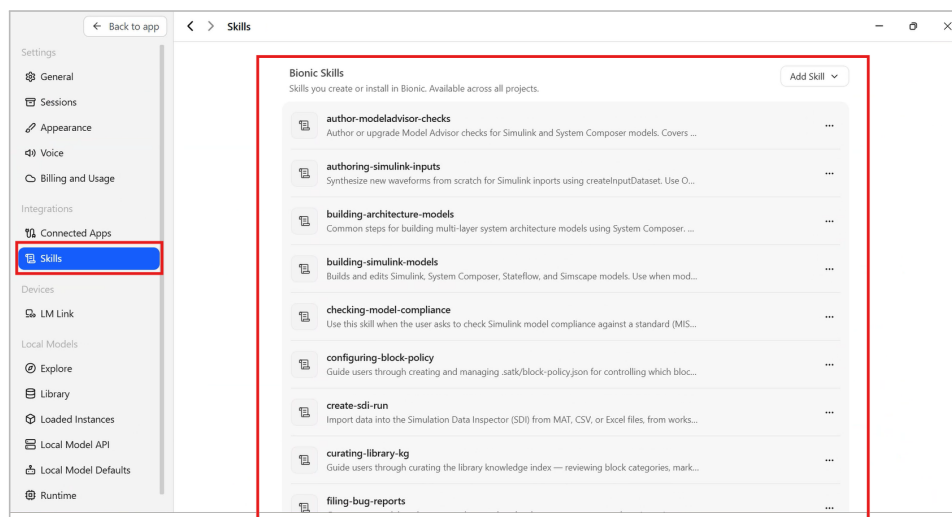


Figure 23: The Bionic Skills list after copying both catalogs

★ Note

If the list is empty, restart LM Studio. If it is still empty, check the folder layout — each skill folder must contain a SKILL.md file.

Step 12. Run Your First Prompt

Your setup is complete. You do not need to start MATLAB yourself — on the first tool call the MCP server launches MATLAB for you. If MATLAB is already open, the agent attaches to that session instead.

1. Click **Back to app** to leave Settings, then click **New Project**. Give the project a name, leave **Allow coding** switched on, pick a folder for the **Local workspace**, and click **Create**.

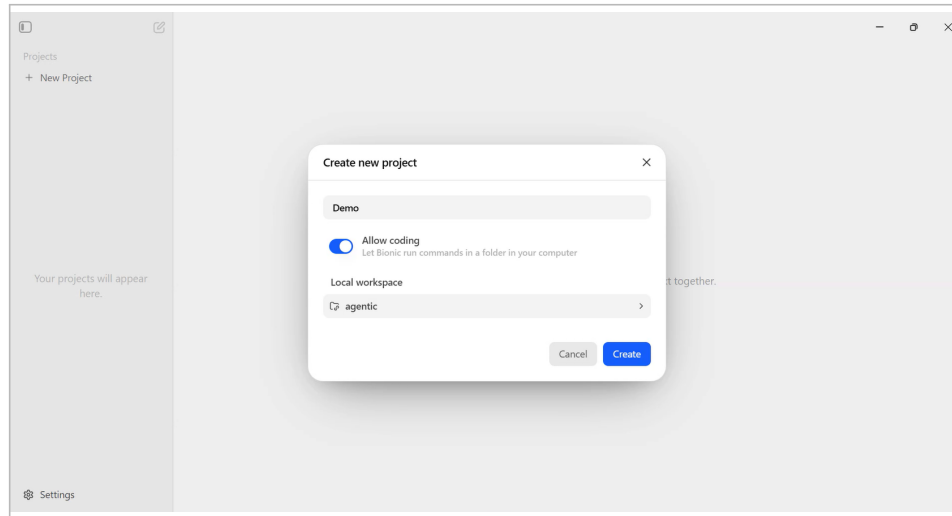


Figure 24: Creating a project with Allow coding switched on

2. In the prompt box, click the model selector and choose the model you downloaded in Step 3.

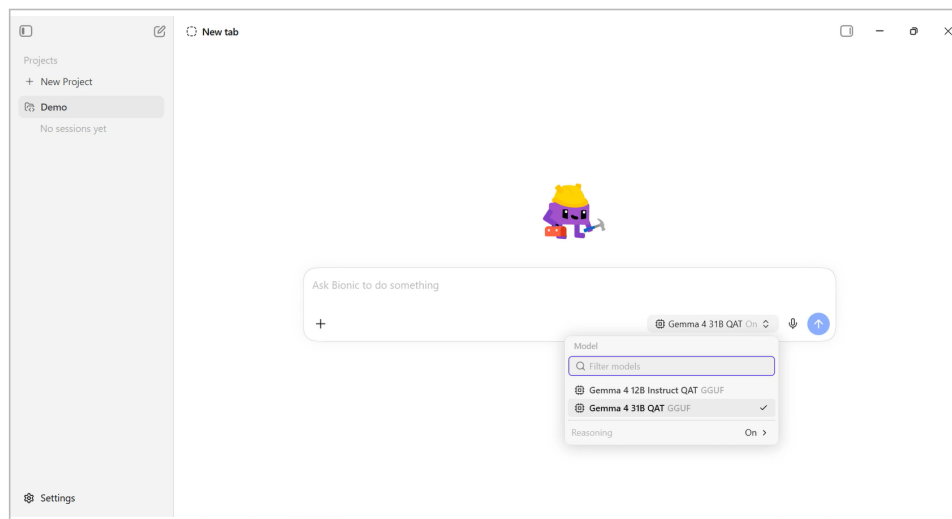


Figure 25: Choosing the model in the prompt box

3. Send this prompt:

What version of MATLAB is running? List the installed toolboxes.

4. Watch the session. The agent shows each tool as it runs — the first is **Called MATLAB MCP Core Server**, and MATLAB starts up alongside it.

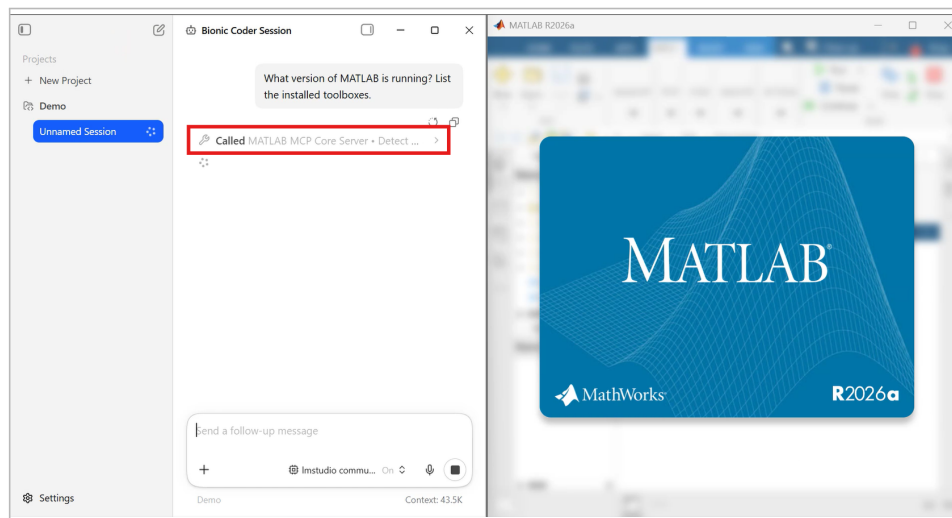


Figure 26: The first tool call, with MATLAB launching automatically

- Wait for the answer. It should report **your** MATLAB release and **your** toolbox list. Expand **Called 1 tool** and **Reasoned** to see which tool the model picked — here it chose `detect_matlab_toolboxes`.

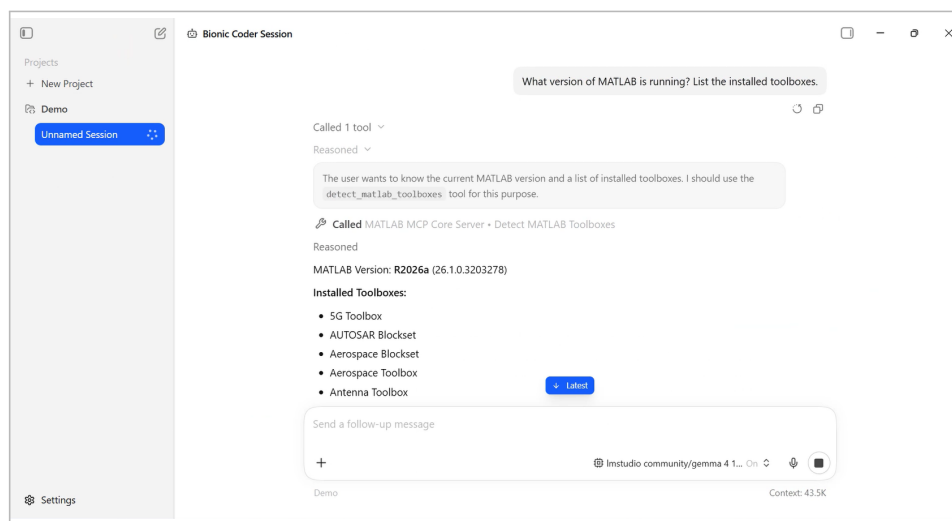


Figure 27: The reply: release and toolbox list read live from MATLAB

★ Note

The **first** prompt is slow, because MATLAB has to start before the tool can return. Later prompts in the same session reuse the running MATLAB and respond much faster.

★ Note

Everything you just ran was offline. To prove it, disconnect from the network and send the prompt again — the model, the MCP server, the tools and the skills are all on your machine. The one exception is MATLAB's own licence check, if yours is a network licence.

Setup Checklist

Run through this list before the workshop:

- ☐ LM Studio is installed and opens without errors
- ☐ A model with the `Tools` capability has finished downloading
- ☐ MATLAB R2026a is installed locally, with Simulink
- ☐ `setupAgenticToolkit` completed and reported **Artifacts installed**
- ☐ **Connected Apps** shows **Connected • 13 tools ready**
- ☐ **Minimum AutoFit context length** is set to 325000
- ☐ **Settings** → **Skills** lists the MATLAB and Simulink skills
- ☐ The first prompt returned **your** MATLAB release and toolbox list

Troubleshooting

Symptom	Most likely cause
MCP server will not connect	The Command path is wrong, or <code>matlab-mcp-server.exe</code> is missing.
Connected, but fewer than 13 tools	The <code>-extension-file</code> argument is missing, or points at a <code>tools.json</code> that does not exist.
Model fails to load	Not enough RAM or VRAM for a 325,000-token context. Lower the value, or use a smaller model.
Answer arrives with no Called row	The model is answering from general knowledge, not from MATLAB. Check that it has the <code>Tools</code> capability, and that Step 7 reads 13 tools ready.
Skills list is empty	The skills were copied one level too deep, or LM Studio needs a restart.
Unrecognized function or variable	<code>setupAgenticToolkit</code> only exists after the <code>.mltbx</code> is installed — redo Step 5.